vector-quantize-pytorch

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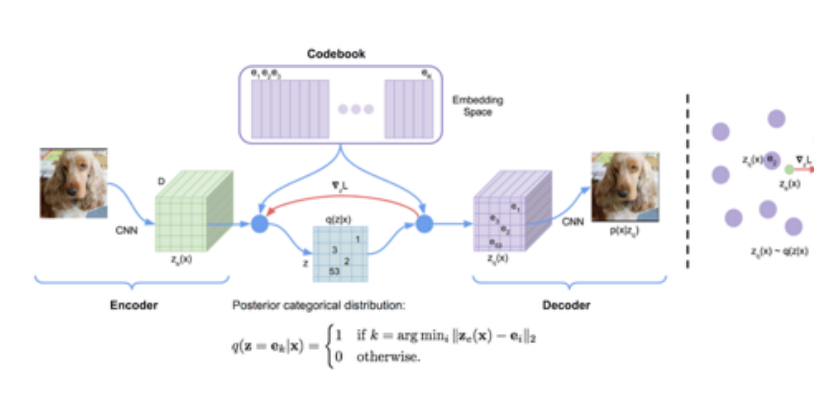
Codebook
Embedding Space
CNN
Encoder
Posterior categorical distribution:
Decoder
$q(z = e_k|x) = \begin{cases} 1 & \text{if } k = \arg\min_j \|z_e(x) - e_j\|_2 \\ 0 & \text{otherwise.} \end{cases}$

Vector Quantization - Pytorch

A vector quantization library originally transcribed from Deepmind's tensorflow implementation, made conveniently into a package. It uses exponential moving averages to update the dictionary.

VQ has been successfully used by Deepmind and OpenAI for high quality generation of images (VQ-VAE-2) and music (Jukebox).

Install

```
$ pip install vector-quantize-pytorch
```

Usage

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_size = 512,     # codebook size
    decay = 0.8,             # the exponential moving average decay, lower means the dictior
    commitment_weight = 1.   # the weight on the commitment loss
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = vq(x) # (1, 1024, 256), (1, 1024), (1)
```

Residual VQ

This paper proposes to use multiple vector quantizers to recursively quantize the residuals of the waveform. You can use this with the `ResidualVQ` class and one extra initialization parameter.

```
import torch
from vector_quantize_pytorch import ResidualVQ

residual_vq = ResidualVQ(
    dim = 256,
    num_quantizers = 8,      # specify number of quantizers
    codebook_size = 1024,    # codebook size
)

x = torch.randn(1, 1024, 256)

quantized, indices, commit_loss = residual_vq(x)
print(quantized.shape, indices.shape, commit_loss.shape)
# (1, 1024, 256), (1, 1024, 8), (1, 8)

# if you need all the codes across the quantization layers, just pass return_all_codes = Tru

quantized, indices, commit_loss, all_codes = residual_vq(x, return_all_codes = True)

# (8, 1, 1024, 256)
```

Furthermore, this paper uses Residual-VQ to construct the RQ-VAE, for generating high resolution images with more compressed codes.

They make two modifications. The first is to share the codebook across all quantizers. The second is to stochastically sample the codes rather than always taking the closest match. You can use both of these features with two extra keyword arguments.

```
import torch
from vector_quantize_pytorch import ResidualVQ

residual_vq = ResidualVQ(
    dim = 256,
    num_quantizers = 8,
    codebook_size = 1024,
    stochastic_sample_codes = True,
    sample_codebook_temp = 0.1,         # temperature for stochastically sampling codes, 0 w
    shared_codebook = True              # whether to share the codebooks for all quantizers
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = residual_vq(x)

# (1, 1024, 256), (1, 1024, 8), (1, 8)
```

A recent paper further proposes to do residual VQ on groups of the feature dimension, showing equivalent results to Encodec while using far fewer codebooks. You can use it by importing `GroupedResidualVQ`

```
import torch
from vector_quantize_pytorch import GroupedResidualVQ

residual_vq = GroupedResidualVQ(
    dim = 256,
    num_quantizers = 8,      # specify number of quantizers
    groups = 2,
    codebook_size = 1024,    # codebook size
)

x = torch.randn(1, 1024, 256)

quantized, indices, commit_loss = residual_vq(x)

# (1, 1024, 256), (2, 1, 1024, 8), (2, 1, 8)
```

Initialization

The SoundStream paper proposes that the codebook should be initialized by the kmeans centroids of the first batch. You can easily turn on this feature with one flag `kmeans_init = True`, for either `VectorQuantize` or `ResidualVQ` class

```
import torch
from vector_quantize_pytorch import ResidualVQ

residual_vq = ResidualVQ(
    dim = 256,
    codebook_size = 256,
    num_quantizers = 4,
    kmeans_init = True,   # set to True
    kmeans_iters = 10     # number of kmeans iterations to calculate the centroids for the c
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = residual_vq(x)

# (1, 1024, 256), (1, 1024, 4), (1, 4)
```

Gradient Computation

VQ-VAEs are traditionally trained with the straight-through estimator (STE). During the backwards pass, the gradient flows *around* the VQ layer rather than *through* it. The rotation trick paper proposes to transform the gradient *through* the VQ layer so the relative angle and magnitude between the input vector and quantized output are encoded into the gradient. You can enable or disable this feature with `rotation_trick=True/False` in the `VectorQuantize` class.

```
from vector_quantize_pytorch import VectorQuantize

vq_layer = VectorQuantize(
    dim = 256,
    codebook_size = 256,
    rotation_trick = True,   # Set to False to use the STE gradient estimator or True to use
)
```

Increasing codebook usage

This repository will contain a few techniques from various papers to combat "dead" codebook entries, which is a common problem when using vector quantizers.

Lower codebook dimension

The Improved VQGAN paper proposes to have the codebook kept in a lower dimension. The encoder values are projected down before being projected back to high dimensional after quantization. You can set this with the `codebook_dim` hyperparameter.

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_size = 256,
    codebook_dim = 16      # paper proposes setting this to 32 or as low as 8 to increase cc
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = vq(x)

# (1, 1024, 256), (1, 1024), (1,)
```

Cosine similarity

The Improved VQGAN paper also proposes to l2 normalize the codes and the encoded vectors, which boils down to using cosine similarity for the distance. They claim enforcing the vectors on a sphere leads to improvements in code usage and downstream reconstruction. You can turn this on by setting `use_cosine_sim = True`

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_size = 256,
    use_cosine_sim = True   # set this to True
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = vq(x)

# (1, 1024, 256), (1, 1024), (1,)
```

Expiring stale codes

Finally, the SoundStream paper has a scheme where they replace codes that have hits below a certain threshold with randomly selected vector from the current batch. You can set this threshold with `threshold_ema_dead_code` keyword.

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_size = 512,
    threshold_ema_dead_code = 2  # should actively replace any codes that have an exponentia
)

x = torch.randn(1, 1024, 256)
quantized, indices, commit_loss = vq(x)

# (1, 1024, 256), (1, 1024), (1,)
```

Orthogonal regularization loss

VQ-VAE / VQ-GAN is quickly gaining popularity. A recent paper proposes that when using vector quantization on images, enforcing the codebook to be orthogonal leads to translation equivariance of the discretized codes, leading to large improvements in downstream text to image generation tasks.

You can use this feature by simply setting the `orthogonal_reg_weight` to be greater than `0`, in which case the orthogonal regularization will be added to the auxiliary loss outputted by the module.

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_size = 256,
    accept_image_fmap = True,                   # set this true to be able to pass in an ima
    orthogonal_reg_weight = 10,                 # in paper, they recommended a value of 10
    orthogonal_reg_max_codes = 128,             # this would randomly sample from the codebc
    orthogonal_reg_active_codes_only = False    # set this to True if you have a very large
)

img_fmap = torch.randn(1, 256, 32, 32)
quantized, indices, loss = vq(img_fmap) # (1, 256, 32, 32), (1, 32, 32), (1,)

# loss now contains the orthogonal regularization loss with the weight as assigned
```

Multi-headed VQ

There has been a number of papers that proposes variants of discrete latent representations with a multi-headed approach (multiple codes per feature). I have decided to offer one variant where the same codebook is used to vector quantize across the input dimension `head` times.

You can also use a more proven approach (memcodes) from NWT paper

```
import torch
from vector_quantize_pytorch import VectorQuantize

vq = VectorQuantize(
    dim = 256,
    codebook_dim = 32,                  # a number of papers have shown smaller codebook dim
    heads = 8,                          # number of heads to vector quantize, codebook share
    separate_codebook_per_head = True,  # whether to have a separate codebook per head. Fals
    codebook_size = 8196,
    accept_image_fmap = True
)

img_fmap = torch.randn(1, 256, 32, 32)
quantized, indices, loss = vq(img_fmap)

# (1, 256, 32, 32), (1, 32, 32, 8), (1,)
```

Random Projection Quantizer

This paper first proposed to use a random projection quantizer for masked speech modeling, where signals are projected with a randomly initialized matrix and then matched with a random initialized codebook. One therefore does not need to learn the quantizer. This technique was used by Google's Universal Speech Model to achieve SOTA for speech-to-text modeling.

USM further proposes to use multiple codebook, and the masked speech modeling with a multi-softmax objective. You can do this easily by setting `num_codebooks` to be greater than 1

```
import torch
from vector_quantize_pytorch import RandomProjectionQuantizer

quantizer = RandomProjectionQuantizer(
    dim = 512,               # input dimensions
    num_codebooks = 16,      # in USM, they used up to 16 for 5% gain
    codebook_dim = 256,      # codebook dimension
    codebook_size = 1024     # codebook size
)

x = torch.randn(1, 1024, 512)
indices = quantizer(x)

# (1, 1024, 16)
```

This repository should also automatically synchronizing the codebooks in a multi-process setting. If somehow it isn't, please open an issue. You can override whether to synchronize codebooks or not by setting `sync_codebook = True | False`

Sim VQ

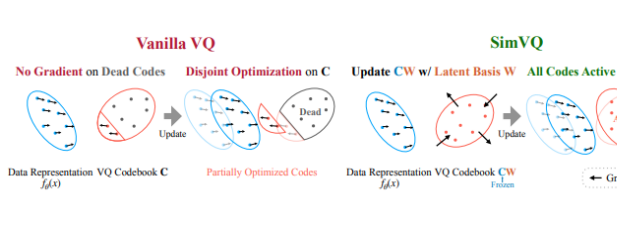
Vanilla VQ
No Gradient on Dead Codes
Disjoint Optimization on C
Update
Dead
Data Representation $f_\theta(x)$
VQ Codebook C
Partially Optimized Codes
SimVQ
Update CW w/ Latent Basis W
All Codes Active on CW
Update
Active
Data Representation $f_\theta(x)$
VQ Codebook CW
Frozen
← Gradient

Figure 1: **Comparison of Vanilla VQ and SimVQ.** (a): (left) Disjoint optimization in Vanilla VQ. Only the nearest codes are updated, resulting in a high percentage of "dead" codes that are not updated. (b): (right) Joint optimization in SimVQ. The entire codebook is updated with a latent basis, ensuring all codes remain active.

A new ICLR 2025 paper proposes a scheme where the codebook is frozen, and the codes are implicitly generated through a linear projection. The authors claim this setup leads to less codebook collapse as well as easier convergence. I have found this to perform even better when paired with rotation trick from Fifty et al., and expanding the linear projection to a small one layer MLP. You can experiment with it as so

Update: hearing mixed results

```
import torch
from vector_quantize_pytorch import SimVQ

sim_vq = SimVQ(
    dim = 512,
    codebook_size = 1024,
    rotation_trick = True  # use rotation trick from Fifty et al.
)

x = torch.randn(1, 1024, 512)
quantized, indices, commit_loss = sim_vq(x)

assert x.shape == quantized.shape
assert torch.allclose(quantized, sim_vq.indices_to_codes(indices), atol = 1e-6)
```

For the residual flavor, just import `ResidualSimVQ` instead

```
import torch
from vector_quantize_pytorch import ResidualSimVQ

residual_sim_vq = ResidualSimVQ(
    dim = 512,
    num_quantizers = 4,
    codebook_size = 1024,
    rotation_trick = True  # use rotation trick from Fifty et al.
)

x = torch.randn(1, 1024, 512)
quantized, indices, commit_loss = residual_sim_vq(x)

assert x.shape == quantized.shape
assert torch.allclose(quantized, residual_sim_vq.get_output_from_indices(indices), atol = 1e
```

Finite Scalar Quantization

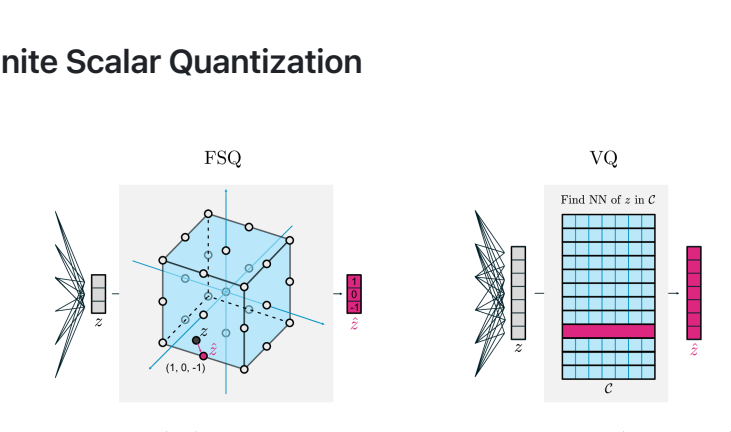

Figure 1: *FSQ (left):* the final encoder layer projects to d dimensions ($d = 3$ shown). We bound each dimension of the encoder output z to L values ($L = 3$ shown), and then round to integers, resulting in the quantized $\hat{z}$, the nearest point in this hypercube. *VQ (right):* The final encoder layer projects to d dimensions ($d = 7$ shown, as d is typically much larger for VQ). The resulting vector z is replaced with the closest vector from the codebook, $\hat{z}$, by nearest neighbor lookup.

	VQ	FSQ
Quantization	argmin_c || z-c ||	round(f(z))
Gradients	Straight Through Estimation (STE)	STE
Auxiliary Losses	Commitment, codebook, entropy loss, ...	N/A
Tricks	EMA on codebook, codebook splitting, projections, ...	N/A
Parameters	Codebook	N/A

This work out of Google Deepmind aims to vastly simplify the way vector quantization is done for generative modeling, removing the need for commitment losses, EMA updating of the codebook, as well as tackle the issues with codebook collapse or insufficient utilization. They simply round each scalar into discrete levels with straight through gradients; the codes become uniform points in a hypercube.

Thanks goes out to @sekstini for porting over this implementation in record time!

```
import torch
from vector_quantize_pytorch import FSQ

quantizer = FSQ(
    levels = [8, 5, 5, 5]
)

x = torch.randn(1, 1024, 4) # 4 since there are 4 levels
xhat, indices = quantizer(x)

# (1, 1024, 4), (1, 1024)

assert torch.all(xhat == quantizer.indices_to_codes(indices))
```

An improvised Residual FSQ, for an attempt to improve audio encoding.

Credit goes to @sekstini for originally incepting the idea here

```
import torch
from vector_quantize_pytorch import ResidualFSQ

residual_fsq = ResidualFSQ(
    dim = 256,
    levels = [8, 5, 5, 3],
    num_quantizers = 8
)

x = torch.randn(1, 1024, 256)

residual_fsq.eval()

quantized, indices = residual_fsq(x)

# (1, 1024, 256), (1, 1024, 8)

quantized_out = residual_fsq.get_output_from_indices(indices)

# (1, 1024, 256)

assert torch.all(quantized == quantized_out)
```

Lookup Free Quantization

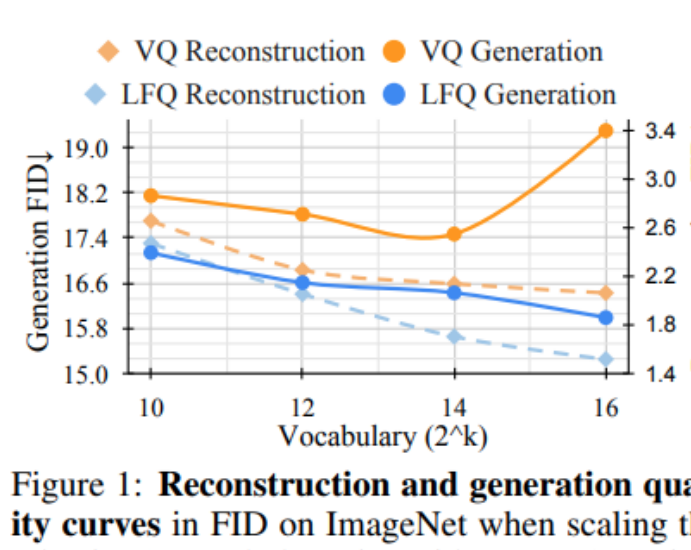

Figure 1: **Reconstruction and generation quality curves** in FID on ImageNet when scaling the tokenizer's vocabulary size with Vector Quantization (VQ) and Lookup-Free Quantization (LFQ). Comparison is done at 128×128 resolution using an MLM with 306-372M parameters.

The research team behind MagViT has released new SOTA results for generative video modeling. A core change between v1 and v2 include a new type of quantization, look-up free quantization (LFQ), which eliminates the codebook and embedding lookup entirely.

This paper presents a simple LFQ quantizer of using independent binary latents. Other implementations of LFQ exist. However, the team shows that MAGVIT-v2 with LFQ significantly improves on the ImageNet benchmark. The differences between LFQ and 2-level FSQ includes entropy regularizations as well as maintained commitment loss.

Developing a more advanced method of LFQ quantization without codebook-lookup could revolutionize generative modeling.

You can use it simply as follows. Will be dogfooded at MagViT2 pytorch port

```
import torch
from vector_quantize_pytorch import LFQ

# you can specify either dim or codebook_size
# if both specified, will be validated against each other

quantizer = LFQ(
    codebook_size = 65536,      # codebook size, must be a power of 2
    dim = 16,                   # this is the input feature dimension, defaults to log2(code
    entropy_loss_weight = 0.1,  # how much weight to place on entropy loss
    diversity_gamma = 1.        # within entropy loss, how much weight to give to diversity
)

image_feats = torch.randn(1, 16, 32, 32)

quantized, indices, entropy_aux_loss = quantizer(image_feats, inv_temperature=100.)  # you m

# (1, 16, 32, 32), (1, 32, 32), ()

assert (quantized == quantizer.indices_to_codes(indices)).all()
```

You can also pass in video features as `(batch, feat, time, height, width)` or sequences as `(batch, seq, feat)`

```
import torch
from vector_quantize_pytorch import LFQ

quantizer = LFQ(
    codebook_size = 65536,
    dim = 16,
    entropy_loss_weight = 0.1,
    diversity_gamma = 1.
)

seq = torch.randn(1, 32, 16)
quantized, *_ = quantizer(seq)

assert seq.shape == quantized.shape

video_feats = torch.randn(1, 16, 10, 32, 32)
quantized, *_ = quantizer(video_feats)

assert video_feats.shape == quantized.shape
```

Or support multiple codebooks

```
import torch
from vector_quantize_pytorch import LFQ

quantizer = LFQ(
    codebook_size = 4096,
    dim = 16,
    num_codebooks = 4  # 4 codebooks, total codebook dimension is log2(4096) * 4
)

image_feats = torch.randn(1, 16, 32, 32)

quantized, indices, entropy_aux_loss = quantizer(image_feats)

# (1, 16, 32, 32), (1, 32, 32, 4), ()

assert image_feats.shape == quantized.shape
assert (quantized == quantizer.indices_to_codes(indices)).all()
```

An improvised Residual LFQ, to see if it can lead to an improvement for audio compression.

```
import torch
from vector_quantize_pytorch import ResidualLFQ

residual_lfq = ResidualLFQ(
    dim = 256,
    codebook_size = 256,
    num_quantizers = 8
)

x = torch.randn(1, 1024, 256)

residual_lfq.eval()

quantized, indices, commit_loss = residual_lfq(x)

# (1, 1024, 256), (1, 1024, 8), (8)

quantized_out = residual_lfq.get_output_from_indices(indices)

# (1, 1024, 256)

assert torch.all(quantized == quantized_out)
```

Latent Quantization

Disentanglement is essential for representation learning as it promotes interpretability, generalization, improved learning, and robustness. It aligns with the goal of capturing meaningful and independent features of the data, facilitating more effective use of learned representations across various applications. For better disentanglement, the challenge is to disentangle underlying variations in a dataset without explicit ground truth information. This work introduces a key inductive bias aimed at encoding and decoding within an organized latent space. The strategy incorporated encompasses discretizing the latent space by assigning discrete code vectors through the utilization of an individual learnable scalar codebook for each dimension. This methodology enables their models to surpass robust prior methods effectively.

Be aware they had to use a very high weight decay for the results in this paper.

```
import torch
from vector_quantize_pytorch import LatentQuantize

# you can specify either dim or codebook_size
# if both specified, will be validated against each other

quantizer = LatentQuantize(
    levels = [5, 5, 8],      # number of levels per codebook dimension
    dim = 16,                   # input dim
    commitment_loss_weight=0.1,
    quantization_loss_weight=0.1,
)

image_feats = torch.randn(1, 16, 32, 32)

quantized, indices, loss = quantizer(image_feats)

# (1, 16, 32, 32), (1, 32, 32), ()

assert image_feats.shape == quantized.shape
assert (quantized == quantizer.indices_to_codes(indices)).all()
```

You can also pass in video features as `(batch, feat, time, height, width)` or sequences as `(batch, seq, feat)`

```
import torch
from vector_quantize_pytorch import LatentQuantize

quantizer = LatentQuantize(
    levels = [5, 5, 8],
    dim = 16,
    commitment_loss_weight=0.1,
    quantization_loss_weight=0.1,
)

seq = torch.randn(1, 32, 16)
quantized, *_ = quantizer(seq)

# (1, 32, 16)

video_feats = torch.randn(1, 16, 10, 32, 32)
quantized, *_ = quantizer(video_feats)

# (1, 16, 10, 32, 32)
```

Or support multiple codebooks

```
import torch
from vector_quantize_pytorch import LatentQuantize

model = LatentQuantize(
    levels = [4, 8, 16],
    dim = 9,
    num_codebooks = 3
)

input_tensor = torch.randn(2, 3, dim)
output_tensor, indices, loss = model(input_tensor)

# (2, 3, 9), (2, 3, 3), ()

assert output_tensor.shape == input_tensor.shape
assert indices.shape == (2, 3, num_codebooks)
assert loss.item() >= 0
```

Citations

```
@misc{oord2018neural,
    title   = {Neural Discrete Representation Learning},
    author  = {Aaron van den Oord and Oriol Vinyals and Koray Kavukcuoglu},
    year    = {2018},
    eprint  = {1711.00937},
    archivePrefix = {arXiv},
```

About

Vector (and Scalar) Quantization, in Pytorch

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+ 6 contributors

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Python 100.0%

```
    primaryClass = {cs.LG}
}
```

```
@misc{zeghidour2021soundstream,
    title   = {SoundStream: An End-to-End Neural Audio Codec},
    author  = {Neil Zeghidour and Alejandro Luebs and Ahmed Omran and Jan Skoglund and Marcc
    year    = {2021},
    eprint  = {2107.03312},
    archivePrefix = {arXiv},
    primaryClass = {cs.SD}
}
```

```
@inproceedings{anonymous2022vectorquantized,
    title   = {Vector-quantized Image Modeling with Improved {VQGAN}},
    author  = {Anonymous},
    booktitle = {Submitted to The Tenth International Conference on Learning Representations
    year    = {2022},
    url     = {https://openreview.net/forum?id=pfNyExj7z2},
    note    = {under review}
}
```

```
@inproceedings{lee2022autoregressive,
    title   = {Autoregressive Image Generation using Residual Quantization},
    author  = {Lee, Doyup and Kim, Chiheon and Kim, Saehoon and Cho, Minsu and Han, Wook-Shi
    booktitle = {Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recog
    pages   = {11523--11532},
    year    = {2022}
}
```

```
@article{Defossez2022HighFN,
    title   = {High Fidelity Neural Audio Compression},
    author  = {Alexandre D'efossez and Jade Copet and Gabriel Synnaeve and Yossi Adi},
    journal = {ArXiv},
    year    = {2022},
    volume  = {abs/2210.13438}
}
```

```
@inproceedings{Chiu2022SelfsupervisedLW,
    title   = {Self-supervised Learning with Random-projection Quantizer for Speech Recognit
    author  = {Chung-Cheng Chiu and James Qin and Yu Zhang and Jiahui Yu and Yonghui Wu},
    booktitle = {International Conference on Machine Learning},
    year    = {2022}
}
```

```
@inproceedings{Zhang2023GoogleUS,
    title   = {Google USM: Scaling Automatic Speech Recognition Beyond 100 Languages},
    author  = {Yu Zhang and Wei Han and James Qin and Yongqiang Wang and Ankur Bapna and Zhe
    year    = {2023}
}
```

```
@inproceedings{Shen2023NaturalSpeech2L,
    title   = {NaturalSpeech 2: Latent Diffusion Models are Natural and Zero-Shot Speech anc
    author  = {Kai Shen and Zeqian Ju and Xu Tan and Yanqing Liu and Yichong Leng and Lei He
    year    = {2023}
}
```

```
@inproceedings{Yang2023HiFiCodecGV,
    title   = {HiFi-Codec: Group-residual Vector quantization for High Fidelity Audio Codec}
    author  = {Dongchao Yang and Songxiang Liu and Rongjie Huang and Jinchuan Tian and Chao
    year    = {2023}
}
```

```
@inproceedings{huh2023improvedvqste,
    title   = {Straightening Out the Straight-Through Estimator: Overcoming Optimization Cha
    author  = {Huh, Minyoung and Cheung, Brian and Agrawal, Pulkit and Isola, Phillip},
    booktitle = {International Conference on Machine Learning},
    year    = {2023},
    organization = {PMLR}
}
```

```
@inproceedings{rogozhnikov2022einops,
    title   = {Einops: Clear and Reliable Tensor Manipulations with Einstein-like Notation},
    author  = {Alex Rogozhnikov},
    booktitle = {International Conference on Learning Representations},
    year    = {2022},
    url     = {https://openreview.net/forum?id=oapKSVM2bcj}
}
```

```
@misc{shin2021translationequivariant,
    title   = {Translation-equivariant Image Quantizer for Bi-directional Image-Text Generat
    author  = {Woncheol Shin and Gyubok Lee and Jiyoung Lee and Joonseok Lee and Edward Choi
    year    = {2021},
    eprint  = {2112.00384},
    archivePrefix = {arXiv},
    primaryClass = {cs.CV}
}
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@misc{mentzer2023finite,
    title   = {Finite Scalar Quantization: VQ-VAE Made Simple},
    author  = {Fabian Mentzer and David Minnen and Eirikur Agustsson and Michael Tschannen},
    year    = {2023},
    eprint  = {2309.15505},
    archivePrefix = {arXiv},
    primaryClass = {cs.CV}
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@misc{yu2023language,
    title   = {Language Model Beats Diffusion -- Tokenizer is Key to Visual Generation},
    author  = {Lijun Yu and José Lezama and Nitesh B. Gundavarapu and Luca Versari and Kihyu
    year    = {2023},
    eprint  = {2310.05737},
    archivePrefix = {arXiv},
    primaryClass = {cs.CV}
}
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@inproceedings{Zhao2024ImageAV,
  title     = {Image and Video Tokenization with Binary Spherical Quantization},
  author    = {Yue Zhao and Yuanjun Xiong and Philipp Krahenbuhl},
  year      = {2024},
  url       = {https://api.semanticscholar.org/CorpusID:270380237}
}
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@misc{hsu2023disentanglement,
    title   = {Disentanglement via Latent Quantization},
    author  = {Kyle Hsu and Will Dorrell and James C. R. Whittington and Jiajun Wu and Chels
    year    = {2023},
    eprint  = {2305.18378},
    archivePrefix = {arXiv},
    primaryClass = {cs.LG}
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@inproceedings{Irie2023SelfOrganisingND,
    title   = {Self-Organising Neural Discrete Representation Learning \`a la Kohonen},
    author  = {Kazuki Irie and R'obert Csord'as and J{\"u}rgen Schmidhuber},
    year    = {2023},
    url     = {https://api.semanticscholar.org/CorpusID:256901024}
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@article{Huijben2024ResidualQW,
    title   = {Residual Quantization with Implicit Neural Codebooks},
    author  = {Iris Huijben and Matthijs Douze and Matthew Muckley and Ruud van Sloun and Ja
    journal = {ArXiv},
    year    = {2024},
    volume  = {abs/2401.14732},
    url     = {https://api.semanticscholar.org/CorpusID:267301189}
}
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@article{Fifty2024Restructuring,
    title   = {Restructuring Vector Quantization with the Rotation Trick},
    author  = {Christopher Fifty, Ronald G. Junkins, Dennis Duan, Aniketh Iyengar, Jerry W.
    journal = {ArXiv},
    year    = {2024},
    volume  = {abs/2410.06424},
    url     = {https://api.semanticscholar.org/CorpusID:273229218}
}
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@inproceedings{Zhu2024AddressingRC,
    title   = {Addressing Representation Collapse in Vector Quantized Models with One Linear
    author  = {Yongxin Zhu and Bocheng Li and Yifei Xin and Linli Xu},
    year    = {2024},
    url     = {https://api.semanticscholar.org/CorpusID:273812459}
}
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@article{Parker2024ScalingTF,
    title   = {Scaling Transformers for Low-Bitrate High-Quality Speech Coding},
    author  = {Julian Parker and Anton Smirnov and Jordi Pons and CJ Carr and Zack Zukowski
    journal = {ArXiv},
    year    = {2024},
    volume  = {abs/2411.19842},
    url     = {https://api.semanticscholar.org/CorpusID:274423406}
}
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